Fail-Closed or Fail-Degraded? Quantifying Forced Assurance Loss in Contested Multi-Hop QKD Networks

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***Abstract*—In multi-hop trusted-relay quantum key distribution (QKD), secret key is a depletable per-link resource, since each hop of a relayed path consumes it. How a network should behave when that key runs short is unsettled. ITU-T Y.3803 permits degradation, allowing a symmetric cipher to replace one-time-pad key relay, and field deployments follow it. A parallel line of security engineering refuses, treating the same transition as a downgrade attack and failing closed instead. Neither position has been measured against the other. This paper quantifies the trade-off. We model a nine-node tactical relay formation with per-link key reservoirs, under an adversary that raises the quantum bit error rate (QBER) until affected links abort key generation. We introduce *forced assurance loss*—the fraction of sessions entitled to an information-theoretically secure key that are served without one—and the *downgrade-forcing cost*, the adversary budget required to induce it. Suppressing links equal in number to the minimum edge cut forces the choice. Fail-closed operation then loses availability, from 0.812 to 0.304. Fail-degraded operation instead preserves availability at 0.973, but serves 69% of entitled traffic below its authorised grade. The forcing cost obeys a two-threshold law: the adversary must both exceed a per-link QBER margin and hold at least the minimum cut; below either threshold its effect is negligible. A post-quantum fallback removes the sensitivity of availability to routing policy, but protects only against quantum-channel attacks. These findings indicate that the assurance-grade composition of served traffic belongs in QKD resilience reporting alongside key-supply continuity. The simulator, validated against four published measurements, is released with all configurations.**

***Index Terms*—Quantum key distribution, trusted-relay networks, key management, post-quantum cryptography, graceful degradation, military communications.**

I. Introduction

A brigade operating in mountainous terrain reaches its rear area through a chain of relay sites. Command circuits are rekeyed every few minutes, imagery from forward observation posts is encrypted at one-time-pad rates on the highest-grade circuits, and logistics traffic is bulk and delay-tolerant. Suppose the network can no longer supply enough quantum key for all of it. If it refuses service, the commander sees an outage and responds to it. If instead it keeps the circuit up but quietly rekeys the imagery link with a computationally secure algorithm rather than an information-theoretically secure one, the commander sees nothing at all, and the loss is invisible until someone audits it. Which of those two failures is preferable is not a settled question, and the two communities that have addressed it disagree.

The setting that makes the question sharp is the trusted-relay network. Beyond the reach of a single quantum channel, keys are extended hop by hop: each intermediate node re-encrypts the end-to-end key under the key it shares with its neighbour, using a one-time pad and an information-theoretically secure authenticator [1]. An n-hop path therefore consumes key on *every* one of its n links, and with free-space per-hop rates of order 10^2 bit/s the key supply, not the existence of a quantum channel, is what binds.

The disagreement. ITU-T Y.3803 cl. 8.2.3 [32] permits degradation outright: where the keys needed for one-time-pad relay are unavailable, a symmetric cipher such as AES may be used instead, and the key manager must record which method it used. Deployments do this [14], and Tysowski *et al.* put both branches under explicit policy control—on exhaustion the key manager "falls back to key derivation if allowed by the policy, or blocks the host" [28]. Against this, a recent layered QKD/post-quantum design treats the same transition as an attack surface, preventing *downgrade attacks* by tearing down the tunnel after a defined interval [10]. The worry is older than the mechanism: an adversary who cannot steal quantum keys will instead deny service precisely in order to *force* the use of classical cryptography [13].

The gap. Both positions are architecturally specified and both are defensible. Neither has been measured. Tysowski *et al.* is a design document that reports no simulation or experiment, and we are aware of no study that quantifies what either branch costs, or what an adversary must spend to force the choice. This matters operationally because the two failure modes are not merely different in kind: they hand the adversary different currencies. Refusing converts attack effort into outage, which is visible; degrading converts it into silent loss of assurance, which is not. Without a measurement, an operator choosing between them is choosing blind.

Contributions. (i) A threat-aware degradation model in which adversarial pressure is expressed in absolute QBER units and a link driven past the security-abort threshold yields *zero* key, with classical-channel denial and QBER-invisible detector attacks as distinct states (§III-B). (ii) *Forced assurance loss*, a metric defined on traffic *entitled* to information-

TABLE I
POSITIONING. *MH* MULTI-HOP TRUSTED RELAY; *KS* KEY SCARCITY; *PC* PRIORITY CLASSES; *CT* CONTESTED; *SW* PRESSURE *swept*; *FB* FALLBACK TO WEAKER CRYPTO; *FA* FORCED-ASSURANCE METRIC.

Work	MH	KS	PC	CT	SW	FB	FA
van Duijn *et al.* [8]	✓	✓	–	–	–	–	–
Mehic *et al.* [7]	✓	✓	✓	○	–	–	–
Cheng *et al.* [27]	✓	✓	✓	–	–	–	–
Johann *et al.* [25]	✓	✓	–	–	–	–	–
Hao *et al.* [24] (RL)	✓	✓	–	–	–	–	–
Cao *et al.* [26] (KoD)	✓	✓	○	–	–	–	–
Tysowski *et al.* [28]	✓	✓	✓	–	–	✓	–
Sanz *et al.* [11]	✓	–	✓	–	–	✓	–
Zhu [12]	○	✓	✓	–	–	✓	○
Spooren *et al.* [10]	✓	–	–	○	–	fail-cl.	–
Makris *et al.* [14]	✓	○	–	○	–	✓	–
Li *et al.* [21]	✓	✓	–	✓	–	–	–
Ghorab *et al.* [22]	✓	✓	–	✓	✓	–	–
Hafeez *et al.* [23]	✓	✓	–	✓	✓	✓	–
This work	✓	✓	✓	✓	✓	✓	✓

theoretic key, and the *downgrade-forcing cost*; we show the latter obeys a two-threshold law in adversary capability and budget (§V-A, §V-D). (iii) Three negative results for key-aware routing once a fallback exists, each with paired confidence intervals under common random numbers (§V-B). (iv) A policy-space audit which finds that an interpretable heuristic beats a searched policy under distribution shift, and explains why (§V-E). (v) A simulator validated against *four* independent published measurements, one of which it fails—identifying an under-parameterised term in our own key accounting (§IV-A).

II. RELATED WORK

Table I positions the work; two columns carry the argument. *FA* is empty in every prior row—the fraction of *entitled* traffic forced below ITS key is not reported anywhere we could find, [12] defining the closest antecedent but neither against swept pressure nor restricted to entitled traffic. And no prior row combines *FB* with *SW*: work that models a fallback does not sweep adversarial pressure, and work that sweeps pressure does not model a fallback. Comparing the two fail modes needs both. Graded delivery itself is established, including four controller-selected assurance levels with a KDF-combined middle tier [11] and architectures that hide the substitution behind an unchanged delivery interface [29]; none of it reports what the substitution costs.

On terminology, *hybrid* in the accepted sense means simultaneous combination via a key combiner, secure if at least one component is unbroken [30]. A QKD→PQC substitution is therefore not hybrid, and we call it a labelled grade downgrade. We also avoid "security level", which already denotes key length in this literature.

III. SYSTEM MODEL

A. Topology, reservoirs and assurance grades

We model (T1) a nine-node formation—a rear node, two relays, two brigade nodes, three forward posts, nine links, one fibre and eight free-space, shown in Fig. 1(a)—and (T2) a chain of depth d with one longer lateral alternate. Fig. 1(b) shows where each control decision sits relative to the key economy.

Reservoirs use the QKDNetSim key-buffer semantics (READY/CHARGING/WARNING/EMPTY on a threshold and a minimum), and generated key exceeding capacity is *discarded*. Capacity, threshold and minimum take that simulator's shipped SECOQC values: 52 428 800, 11 324 620 and 1 048 576 bits. The minimum doubles as an *authentication reserve*: SECOQC held a 2 MiB floor purely to guarantee key for future classical-channel authentication [2]. No vendor publishes a storage capacity, so capacity is swept.

Sessions are served at one of four grades: G_1 one-time pad (demand = payload bits, ITS payload, per Shannon [6]); G_2 QKD-keyed symmetric (256 bit, ITS-*delivered* key); G_3 hybrid QKD $\oplus$ ML-KEM-768 (128 bit plus KEM, via an approved domain-separated combiner); G_4 ML-KEM-768 only (no quantum key). Each class carries an *authorised floor*. When the key economy cannot meet it the controller either blocks (FAIL_CLOSED) or serves below it and records an auditable floor violation (FAIL_DEGRADED).

B. Link model and adversary

A two-state model giving a degraded link a fixed fraction of nominal rate is indefensible under attack: a denial-of-service adversary acts on the quantum channel specifically to raise noise above the security threshold, and any perturbation strong enough to do so disrupts key generation entirely [5]. We use four states: NOMINAL (R_0); DEGRADED ($R_0\rho(e)$); ABORT (0, for $e \geq e_{\mathrm{abort}}$); OUTAGE (0; cut, LOS loss, *or classical-channel denial*), with

$$\rho(e) = \left[\frac{\max(0,\, 1 - fH_2(e) - H_2(e))}{1 - fH_2(e_0) - H_2(e_0)}\right]^{\gamma}, \qquad (1)$$

$f = 1.16$ and $e_0 = 0.03$ [3]. We report $\gamma = 1$ (asymptotic BB84 form, following [4]) and $\gamma = 3.4$, calibrated to the only measured degraded-link datapoint we found: a deliberately misaligned entanglement link at QBER 5–7% delivered 0.25 kbps against 2–3 kbps nominal [3]. These give 50% and 10% at QBER 6%; **we publish the disagreement as a sensitivity rather than resolve it** (§V-D shows it barely matters). $e_{\mathrm{abort}} = 0.075$ is an engineering value—the deployed distillation engine in [3] provides coding rates only below 3.5, 5.5 and 7.5%—and 0.11 is reported as a sensitivity. Note that realistic error correction *lowers* the ITS zero from 11.0% to 9.9% at $f = 1.16$.

Two failure modes bypass QBER entirely and are separate states, which we state as a strength: denial of the classical post-processing channel leaves the quantum channel pristine while halting key production [7]; and detector blinding by tailored bright illumination makes it possible to "tracelessly acquire the full secret key" [20], from which it follows that such an attack is invisible to any QBER-driven model including ours.

The adversary attacks availability to force a downgrade [13]: it selects k links and injects excess QBER Δe. Expressing pressure in absolute QBER units makes it checkable. We

(a) formation, and the cut that severs it

(b) control loop and the two attack surfaces

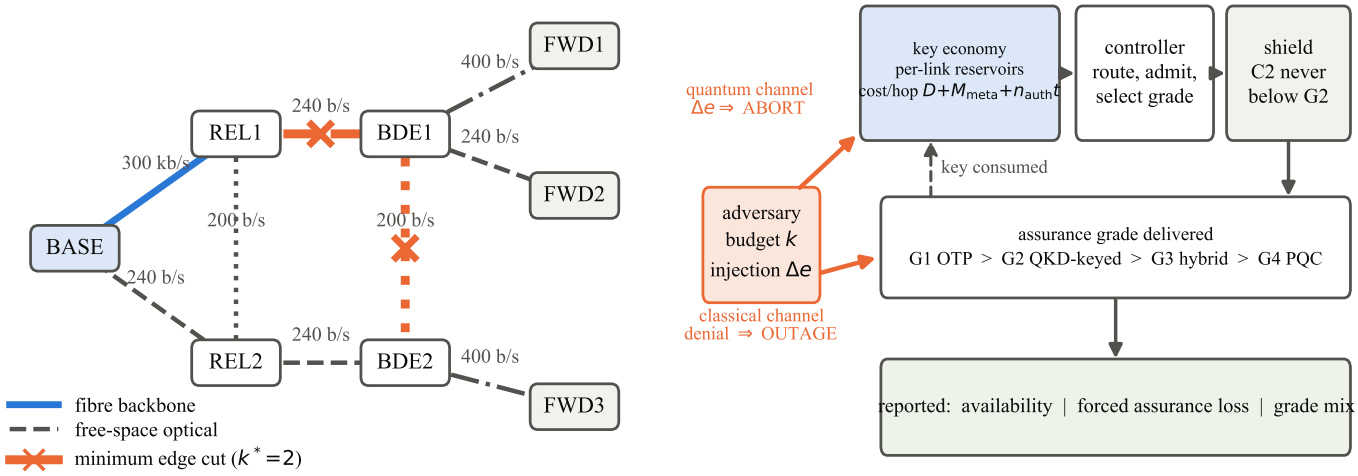


Fig. 1. System architecture. (a) The nine-node formation, with per-link secret key rates and the minimum edge cut marked: suppressing those two links severs quantum key to two of the three forward posts, which is why $k^* = 2$ here. (b) The control loop. Each session is routed, admitted and assigned an assurance grade; a hard shield forbids command traffic below G_2; consumption debits the per-link reservoirs. The adversary has two distinct entry points, and they are not equivalent—raising QBER on the quantum channel aborts key generation while leaving the classical channel intact, so the post-quantum grade still works, whereas denying the classical channel removes that grade as well.

report two targeting models, greedy by edge betweenness and *minimum edge cut*—the honest worst case.

C. Key accounting and traffic

Relaying D bits of key across one hop costs $D + M_{\text{meta}} + n_{\text{auth}}t$, with metadata 240 bit (Y.3803 relays a *key file*, not a bare key) and authentication tag $t = 127$ bit [16]. Wegman–Carter authentication is charged per message and grows only *logarithmically* in message length [17], so the overhead is a function of block size: at $D = 256$ the total is 2.93 times the delivered key, and at $D = 1$ Mbit it is 1.000. Setting the authentication terms to zero recovers the bare $D{\cdot}n$ law used throughout the routing literature, which we treat as an ablation.

The count n_{auth} of authenticated messages per key delivery is the one term we could not ground in a specification. We initially set it to 2, from the observation that mutual authentication consumes two independent one-time keys per round. §IV-A shows this is a lower bound rather than a measurement: published relay overhead implies ≈ 7 and an application-level tunnel ≈ 40. We therefore report it as a swept parameter and re-check the headline across the range.

Demands come from measured deployments. C2: 2048 bit, floor G_2, 5 s, $\lambda = 0.05\,\text{s}^{-1}$ (256 bit/min per encryptor is the modal deployed unit [19]). ISR: 144 000 bit, floor G_1, 30 s, $\lambda = 0.02\,\text{s}^{-1}$—a 60 s session of the 2.4 kbps real one-time-pad voice run for 31 months on a 46-node metropolitan QKD network [18]. LOG: 320 000 bit, floor G_3, 120 s, $\lambda = 0.03\,\text{s}^{-1}$, from the same network's measured 320 kbps file transfer.

IV. Experimental Setup

Discrete-event simulation at 1 s resolution, 24 h horizon with the first 12 h discarded. The warm-up is itself a finding: with reservoirs initialised full, free-space stores take ∼12 h to fall to the authentication reserve, and half-run metrics agree only thereafter. A 4 h run leaves them at 59% of capacity and measures the initial fill, not the key economy.

Policies use *common random numbers*—identical arrival streams and channel realisations per seed—so comparisons are paired; intervals are paired bootstraps (20 000 resamples). Twelve seeds for the main comparisons, six for sweeps.

A. Validation, and why not against another simulator

Cross-validating against an established quantum-network simulator would be the natural check and we could not do it, for a reason worth stating precisely. Source inspection shows SeQUeNCe implements point-to-point BB84 with Cascade and has no trusted-relay or key-pool model, QuISP models entanglement distribution and has no key pools, and NetSquid has neither while distributing package and documentation behind registration under a non-OSI licence. *A key-economy simulator cannot be validated against tools that do not model a key economy.* QKDNetSim is the sole on-target candidate and we adopt its buffer semantics verbatim, so our reservoir is a citable object rather than an invention; but its routing is self-described as an early implementation, *admission control is absent*—one of the two mechanisms compared here—and QKDNetSim+ reports that its nodes did not adequately process received key material and that its crypto handler malfunctioned [31], defects in exactly the layer a cross-validation would exercise. We list it as future work.

We instead constrain the model from four independent directions, using three deployments. A 43-check gate covering key conservation, reservoir bounds and transitions, the rate cliff and the degenerate cases in both fail modes must pass

TABLE II
VALIDATION AGAINST PUBLISHED MEASUREMENTS. TWO CHECKS DISAGREE, IN THE SAME DIRECTION, AND THE DISAGREEMENT IDENTIFIES AN UNDER-PARAMETERISED TERM.

	Quantity	Measured	Model
V1	key-store drain [2]	3584 s	**3584 s**
V2	relay overhead [15]	$3.69\times D$	$2.23\times D$
V3	auth. share [2]	91.2%	65.9%
V4	OTP voice pairs [18]	11	1.2–12.6

TABLE III
AVAILABILITY AND FORCED ASSURANCE LOSS VERSUS ADVERSARY BUDGET (12 SEEDS, KEY-AWARE ROUTING, $\Delta e = 0.05$, BETWEENNESS TARGETING).

Mode	k	Avail.	Blocked	G_2	G_4	Forced
CLOSED	0	0.8124	0.1876	0.590	0.000	**0.000**
CLOSED	1	0.8264	0.1736	0.564	0.000	**0.000**
CLOSED	2	**0.3039**	0.6961	0.508	0.000	**0.000**
CLOSED	5	**0.0000**	1.0000	—	—	—
DEGR.	0	0.9733	0.0267	0.659	0.000	0.166
DEGR.	1	0.9733	0.0267	0.607	0.029	0.157
DEGR.	2	0.9733	0.0267	0.174	**0.675**	**0.690**
DEGR.	5	0.9733	0.0267	0.000	**1.000**	**1.000**

before any result is reported; on top of that we reproduce the following.

V1 reproduces the SECOQC rerouting experiment, in which an 8192 bit/s one-time-pad application drained 3.5 MiB of a 5.5 MiB key store in about an hour: we obtain 3584 s against a reference of 3584 s. V4 checks the one-time-pad demand law and passes untuned—eleven simultaneous bidirectional 2.4 kbps voice pairs need 52.8 kbps, inside the 6–60.5 kbps range measured on that network.

V2 and V3 both fail, and they fail the same way. Cambridge distributed 1000 global keys for about 180 kbyte of link key, 3.69 times the ∼400-bit key delivered, where we predict 2.23; SECOQC's tunnel spent 5824 bit per rekey round of which only 512 bit was cipher key—91.2% authentication—where we predict 65.9%. Solving each for the number of authenticated messages per key delivery gives $n_{\mathrm{auth}} \approx 7$ for a relay hop and ≈ 40 for an application-level tunnel, against the 2 we assumed. **Our $n_{\mathrm{auth}} = 2$ is a lower bound, not a measurement, and our per-hop cost is correspondingly optimistic.**

This is a result rather than an embarrassment. It fixes the *direction* of our bias, which is what a reader needs to use the numbers; it strengthens the accounting finding of §III-C, since calibrating to Cambridge grows the understatement in the bare $D{\cdot}n$ law from 2.93× to 5.41× at 256-bit blocks; and it forced the headline re-check at the calibrated value reported in §V-D, which the headline survives.

V. RESULTS

A. The fail-closed/fail-degraded exchange rate

The transition is a cliff (Table III, Fig. 2). Suppressing 2 of 9 links collapses fail-closed availability from 0.812 to 0.304, while fail-degraded operation holds availability at 0.973 and converts the same pressure into 69.0% forced assurance loss, post-quantum-only service rising from 2.9% to 67.5%. At $k = 5$ both endpoints are total: complete denial, or complete assurance loss at undiminished availability.

The general law, not the instance. The value 2 is a property of the cut structure, not of our formation: k^* equals the *minimum edge cut* separating the rear node from the forward posts. Here two links are the only quantum-viable route into one brigade node, so the cut is 2 and availability floors at one third. §V-C confirms the law on a chain where the cut is 2 at *every* depth.

The authorised policy is infeasible even unattacked, and we say so. At $k = 0$, 16.6% of entitled traffic is already forced below its floor: at the free-space rates cited, a one-time-pad floor for a 144 kbit session is not fundable. Either the rate must rise or the floor must relax to periodic symmetric rekeying.

B. Three negative results for key-aware routing

(1) The service metric becomes degenerate. Under fail-degraded operation the paired availability difference between all three policies is 0.0000 with a 95% CI of *zero width*, because the fallback succeeds wherever a classical path exists. Availability is then a property of connectivity, not of key-awareness: *the metric the key-aware-routing literature reports stops responding to the thing it measures*. Routing still moves key expenditure significantly (KAR +27.0 Mbit, CI $[+25.3, +28.9]$; floors −38.1 Mbit, CI $[-40.5, -35.7]$).

(2) Key-aware routing is worse. Under fail-closed operation KAR loses 2.7 points of availability to shortest path (-0.0268, CI $[-0.0293, -0.0243]$). The mechanism is the hop multiplier: KAR selects marginally longer paths (3.086 vs 3.062 hops) and every extra hop consumes key on an *additional* link, raising expenditure 6.0% and starving later sessions. Sweeping the hop penalty over four orders of magnitude does not repair it: at penalty 10^3 KAR converges to shortest-path behaviour (3.065 hops, 440.0 Mbit) rather than beating it. We scope the claim to its three joint conditions—binding scarcity at steady state, fail-closed semantics, and a hop-multiplied cost with authentication charged. Where prior work reports 25–40% improvements [9] it measures blocking under different scarcity and without a fallback.

(3) Reservation floors are a Pareto trade. Class-aware floors convert availability into assurance: $0.839 \to 0.509$ in exchange for forced loss $0.357 \to 0.0000$. Every session served carries an ITS-delivered key.

C. Downgrade-forcing cost versus depth

Measured on *forced* degradation (nominal loss is 0.000 at every depth, so every point is attributable to the attack), a greedy centrality-guided adversary requires $k^* = d + 1$, growing linearly with depth, because it exhausts the already-severed primary path before touching the alternate. A minimum-cut adversary requires **exactly two links at every depth** (Fig. 3). At $d = 6$ that is a 3.5× overstatement of resilience if only the greedy adversary is reported. *Adding relay depth buys no protection against a competent adversary.*

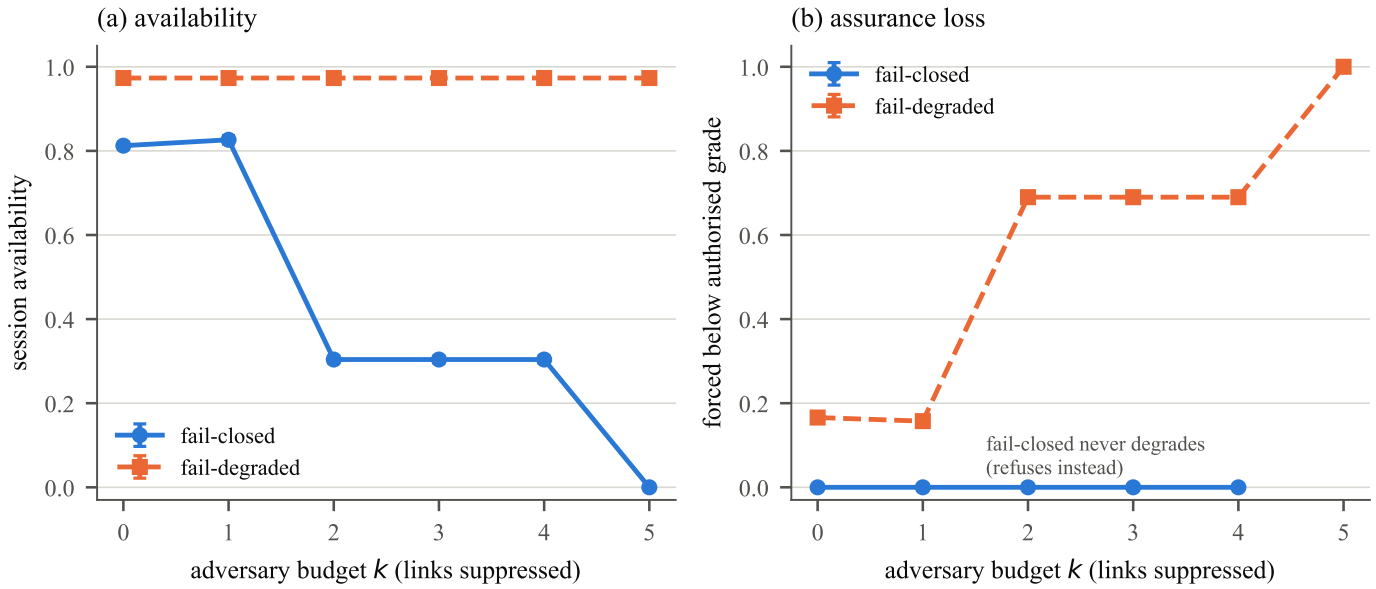


Fig. 2. Availability (a) and forced assurance loss (b) versus adversary budget k, for fail-closed and fail-degraded operation. Error bars: SD over 12 seeds. Fail-closed cannot lose assurance by construction—it refuses instead.

TABLE IV

POLICY COMPARISON, PAIRED BOOTSTRAP OVER COMMON SEEDS (20 000 RESAMPLES, 12 SEEDS), $k = 0$. SP: SHORTEST PATH. KAR: KEY-AWARE. CAF: CLASS-AWARE FLOORS. SET FULL WIDTH BECAUSE THE PAIRED INTERVALS ARE THE RESULT—REPORTING THEM ONLY IN PROSE WOULD LEAVE A READER UNABLE TO SEE THAT THE FAIL-DEGRADED DIFFERENCES ARE EXACTLY ZERO WHILE KAR'S FAIL-CLOSED DEFICIT IS NOT.

Mode	Policy	Avail.	Δ vs SP [95% CI]	Forced	Key (Mbit)
DEGR.	SP	0.9733	reference	0.166	436.8
DEGR.	KAR	0.9733	+0.0000 [0.0000, 0.0000]	0.157	463.7
DEGR.	KAR+CAF	0.9733	+0.0000 [0.0000, 0.0000]	0.166	398.7
CLOSED	SP	**0.8392**	reference	0.3569	437.8
CLOSED	KAR	**0.8124**	**−0.0268 [−0.0293, −0.0243]**	0.3687	464.1
CLOSED	KAR+CAF	**0.5092**	**−0.3299 [−0.3333, −0.3264]**	**0.0000**	399.3

Under fail-degraded operation the three policies coincide on availability *exactly*: the confidence interval has zero width because the post-quantum fallback succeeds on every session for which a classical path exists, in every seed. Mean path length is 3.060–3.128 hops throughout and is omitted for space.

D. Sensitivity and the two-threshold law

The parameter we worried about most turns out not to matter. The five-fold calibration ambiguity in γ changes almost nothing: fail-closed availability at $k = 2$ is 0.302 at *both* γ values for $e_{\text{abort}} = 0.075$. The abort threshold decides everything, and Fig. 4 shows why. At $e_{\text{abort}} = 0.075$ availability holds near 0.80 for $\Delta e \leq 0.04$ and falls to 0.302 at $\Delta e = 0.05$; at $e_{\text{abort}} = 0.11$ the same collapse waits until $\Delta e = 0.08$. The cliff lands *exactly* at $\Delta e \geq e_{\text{abort}} - e_0$ in both cases, and above it the outcome saturates immediately regardless of margin. Hence

> the downgrade-forcing cost has two necessary thresholds: a per-link capability $\Delta e \geq e_{\text{abort}} - e_0$, and a budget $k \geq$ the minimum edge cut.

This is actionable in an unexpected direction. Raising e_{abort} from 0.075 to 0.11 raises the injection an adversary must achieve by **78%**, obtainable through better reconciliation efficiency rather than higher key rate.

The fallback needs the classical channel. Fail-degraded availability is invariant under QBER injection because a link in ABORT still carries a pristine classical channel and ML-KEM needs only that. Under classical-channel denial availability falls from 0.970 to **0.313**—only 1.6 points above fail-closed's 0.296—and forced loss is zero because severed posts are blocked outright rather than downgraded. *The entire advantage of degrading gracefully is conditional on the classical channel surviving.* A defender who hardens the quantum channel and leaves the classical one exposed has bought almost nothing.

The headline survives the authentication recalibration. Since §IV-A shows our $n_{\text{auth}} = 2$ is optimistic, and a larger value tightens scarcity, the result has to be re-checked at the calibrated setting. Sweeping $n_{\text{auth}} \in \{2, 7, 20, 40\}$—a 7.4-fold range in per-hop cost for a 256-bit block, 750 to 5576 bit—moves fail-closed availability at $k = 2$ only from 0.302 to 0.281 and leaves fail-degraded forced loss at 0.678. Aggregate key consumption barely moves either, and the reason is instructive: one-time-pad ISR sessions at 144 kbit

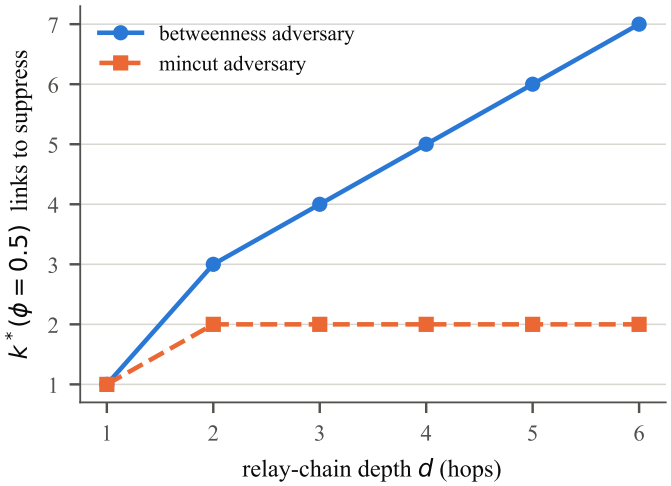


Fig. 3. Downgrade-forcing cost $k^*(\phi{=}0.5)$ versus relay-chain depth for a greedy centrality-guided and an optimal minimum-cut adversary.

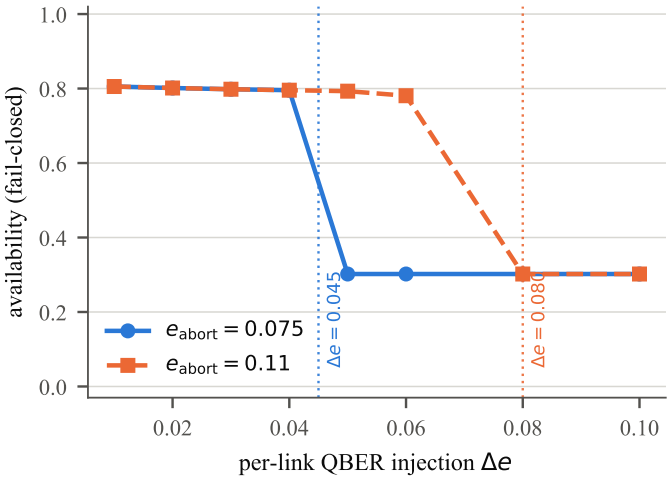


Fig. 4. The two-threshold law. Dotted lines mark the *predicted* capability threshold $\Delta e = e_{\text{abort}} - e_0$; each curve turns at its own line and then saturates. Fail-degraded forced assurance loss mirrors this exactly, rising $0.000 \to 0.677$ at the same two points.

dominate the key budget, so a term of at most a few thousand bits per hop is lost in the noise. *The authentication correction matters for capacity planning in small-block deployments and not for this paper's headline*, which is the same workload-dependence reported in §**??**.

E. Policy-space audit: where learning helps, and why not here

Why we considered learning. Grade selection under key scarcity has the shape that reinforcement learning is usually good at. Decisions are sequential, they consume a shared resource, and their cost is deferred—serving one session at a high grade now denies key to several later ones, across a state space too large to reason about by hand. Two things nevertheless argued against a conventional learned controller. Learned *routing* for key-scarce QKD networks is already well covered, including deep RL with graph attention that generalises across topologies [24], and learned admission control is saturated in classical networking; a routing or admission claim was therefore unavailable to us. More importantly, a learned policy that merely posts a better number in a simulator we wrote ourselves would be unfalsifiable. So we inverted the question. Rather than asking whether learning can beat our heuristic, we asked how much headroom the heuristic leaves, and what a search discovers that we had not thought of.

Method, and why not a network. We search a 12-cell state→grade lookup table—2.4×10^8 policies—by the cross-entropy method, scoring every candidate with full simulation rollouts. Full rollouts matter: because consumption is deferred, a myopic learner would need an invented opportunity-cost term, whereas rollouts price the sequential economy exactly. A deterministic shield masks any action that would place command traffic below G_2. We state plainly that the safety guarantee comes from the shield, which can be checked by reading it, and that search only explores inside it.

Why the heuristic won. At full horizon the hand-designed policy reaches 104.8% of the best searched policy and wins on *all seven* out-of-distribution conditions—secret-key rate $\times 0.25$, $\times 0.5$, $\times 2$, adversary $k \in \{2, 4\}$, doubled outage rate, doubled demand. We report this because we committed to reporting it either way. But calling the null weight-robust would be wrong, and checking showed as much: the two policies do not order componentwise. The searched policy genuinely wins availability, 0.9726 against 0.8123, and pays for it with 39.5% forced downgrades. Writing $J = \text{avail} - \lambda \cdot \text{viol} + w \cdot \text{ITS}$, the heuristic wins precisely when $\lambda > 0.370$ at $w = 0.5$ (0.406 at $w = 0$, 0.264 at $w = 2$).

> The useful output is therefore not a winner but a boundary: *is one forced downgrade of entitled traffic worth more or less than 0.37 of a served session?* Above that price the simple heuristic is correct and no learned policy is needed; below it, the network should run closer to what the search found and accept the assurance loss.

Locating λ^* required the search, because it depends on the achievable trade-off surface—a property of the key economy rather than of the objective—so it cannot be obtained by reasoning about weights alone.

Why the shift occurred: the reservoir transient. The searched policy *beat* the heuristic on the 6 h training horizon (J 1.4591 against 1.2978) and *lost* at 24 h. The cause is specific and it is not overfitting in the usual sense. Reservoirs are initialised full and take roughly 12 h to drain to the authentication reserve (§**??**), so a 6 h training window samples only the key-abundant regime. The agent optimised a state distribution the deployed system leaves behind. This generalises beyond QKD: *in any environment with a depletable shared resource, the training horizon must span the depletion timescale, or the policy is fitted to a transient*. The failure is invisible from training curves, which is what makes it dangerous—the learned policy looked better throughout training and was worse in deployment. We note that none of the learned-routing papers for QKD networks we surveyed reports its training horizon

relative to reservoir dynamics, so this is a checkable question to put to that literature rather than a hypothetical.

What removing the shield revealed. The unshielded ablation scored highest of all, and it did so by cutting key consumption from 464 to 42 Mbit—an $11\times$ reduction—while keeping 100% of traffic on ITS-*delivered* key. It achieved this by serving one-time-pad-entitled ISR traffic at G_2 (256 bit) instead of G_1 (144 000 bit): still delivered by QKD, but a violation of that class's authorised floor. The policy satisfied the letter of the objective while defeating its intent, and our shield did not stop it because the shield constrained only command traffic. Command-below-ITS is 0.0000 in both arms, so the ablation does *not* measure the price of the command guarantee—it demonstrates that an assurance policy specifying a floor for one class is gameable, and that an objective pricing floor violations below key savings will be gamed. Per-class floors must be specified *and* priced.

Why interpretability was designed in rather than recovered. We searched a tabular policy instead of training a network and distilling it, so there is no distillation step and no fidelity loss to report; the resulting policy is twelve rules an operator can read. That choice also exposed a trap worth passing on. The *raw* searched table assigns G_4 to command traffic in one reservoir state, while the *effective* post-shield policy assigns G_1. A reader shown the raw table would conclude the safety guarantee had been violated when it never was. Learned policies operating behind a shield should be reported after projection, not before.

Why we regard this as an AI result. Three of the findings above are claims about learned control, not about quantum networks, and none depends on the QKD setting: a training horizon shorter than a resource's depletion timescale teaches the wrong regime; a safety shield covering one traffic class is gameable through the classes it omits; and an interpretable-by-construction policy class avoids the fidelity loss that post-hoc distillation incurs. The broader point is methodological. A learned controller that wins in a simulator its authors also wrote establishes very little. A learned controller that *loses*, together with a mechanism explaining why and a price at which the conclusion would reverse, is a result that transfers.

VI. Discussion and Limitations

When key runs out, choosing between refusing and degrading chooses which resource the adversary takes, and the exchange rate is steep. Limitations, plainly: per-link key rates are parameterised inputs, not simulated photonics; the measured and asymptotic degradation calibrations disagree fivefold and we report both rather than resolve them (§V-D shows the headline is insensitive to this); relay nodes are assumed trusted, with compromise modelled only as an availability event; detector blinding raises no QBER and is invisible to this model by construction; one formation and one chain family; unmeasured parameters (free-space outage statistics, reservoir capacity) are swept rather than assumed; and the policy audit used a modest search budget.

Future work: a grade attribute carried on the delivered key so a consumer can enforce policy on it, which the current REST delivery interface [33] does not provide; terrain-correlated, spatially correlated outage on free-space relay chains, which every model we found treats as independent; and a per-class downgrade-authorisation language with commander-visible reporting.

VII. Conclusion

Standards permit a QKD network to fall back to weaker key supply; a parallel line of engineering forbids it as a downgrade attack. We measured the trade-off. Suppressing links equal to the minimum edge cut either collapses availability from 0.812 to 0.304 or holds it at 0.973 while forcing 69.0% of entitled traffic below ITS protection, and the forcing cost obeys a two-threshold law in capability and budget. Along the way, a post-quantum fallback makes the key-aware-routing service metric degenerate, key-aware routing costs 2.7 points of availability untunably, and fail-degraded operation is immune only to quantum-channel attacks. We argue that the assurance-grade composition of *served* traffic belongs in QKD resilience reporting alongside key-supply continuity.

One result deserves separating from the networking conclusions because it is not about quantum networks. A hand-designed, twelve-rule interpretable policy outperformed a searched policy on every out-of-distribution condition we tested, and the reason was diagnostic rather than incidental: the search had been trained inside the interval during which the key reservoirs were still draining from full, so it optimised a state distribution the deployed system leaves behind. The same experiment, run with the safety shield removed, then cut key consumption elevenfold by downgrading traffic the shield did not happen to cover—satisfying the objective while defeating its intent. We take two transferable lessons. Where a learned controller shares a depletable resource, the training horizon must span the resource's depletion timescale, or the policy is fitted to a transient that training curves will not reveal. And a safety constraint written for one traffic class will be circumvented through the classes it omits, so assurance floors must be specified and priced for every class, not only the most important one. Neither lesson depends on the setting being quantum, and we would expect both to recur wherever interpretable heuristics and learned policies are compared under resource depletion.

Reproducibility

Simulator, configurations, the 43-check gate and all figure/table scripts are at [[URL]]. Twenty-four-hour horizons, 12 h warm-up discarded, common random numbers, paired bootstraps (20 000 resamples). The full set runs in $\sim$30 min on eight CPU cores; no GPU. ITU-T/ETSI digital products are not redistributed; a download script is provided.

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